FORMATIVE ASSESSMENT – 1 (FA-1)

MATHEMATICS • CLASS 9 • GENERATING NEW PROBLEMS

Problem 1

Express the recurring decimal **0.3777... (0.37̄)** in the fractional form **p / q**, where *p* and *q* are integers and $q \neq 0$. Show complete step-by-step simplification.

Problem 2

Simplify the expression by rationalizing the denominator: **5 / (√7 − √2)**. Write the final answer in simplest radical form.

Problem 3

Find the value of the constant **k** if **(x − 2)** is a factor of the cubic polynomial $\mathbf{p(x) = 2x^3 - 6x^2 + 5x + k}$ using the Factor Theorem.

Problem 4

Without actually calculating the individual cubes, find the numerical value of $\mathbf{(32)^3 + (-18)^3 + (-14)^3}$ by applying a suitable algebraic identity.

Problem 5

Check whether **g(x) = 2x − 1** is a factor of $\mathbf{q(x) = 4x^3 - 4x^2 + x - 1}$ by using the Remainder Theorem method.

Note for Students: Formulate your own variations for each problem type by modifying the coefficients and show the complete solution with justification.